Indian Institute of Technology Gandhinagar Proposal for a New Course	
Course No.	EE 313
Proposed Course Title	Communication Systems
Expected Study Load (Hours)	120
Proposed Credits	3
Proposed L-T-P	2-1-0
Potential Instructors	EE faculty
Proposer	Ravi Hegde, Arup Lal Chakraborty
Expected frequency of offering this course	Each year

Course contents:

Components of a communication link, channel signal to noise ratio, channel capacity; Modulation and demodulation; Analog and digital communication links; Review of signals and spectra, signal vector space; Review of random processes and power spectral density; Analog communication Baseband vs carrier communications, Amplitude modulation, QAM, Angle modulation (PM, FM); Narrowband and wide-band FM. Transmission bandwidth analysis, signal generation and detection; Frequency division multiplexing; Pulse code modulation, sampling theorem, aliasing, analog-to-digital conversion and quantisation noise; Uniform and non uniform quantisation, time division multiplexing; delta modulation, DPCM; Digital communications basics, Baseband and carrier-modulated digital, various line codes and their PSD, pulse shaping, inter-symbol interference, Nyquist ISI criterion, Duo-binary signalling and differential encoding; Noise analysis - Gaussian, Rayleigh and Rician distributions; Eye diagrams, timing extraction, Detection - non-coherent and coherent, BER analysis of PCM; Digital bandpass modulation, ASK, PSK, FSK, BER analysis.

Texts/References:

- Lathi, B. P. and Ding, Z., Modern Digital and Analog Communication Systems, 4th edition, Oxford University Press, 2009. [Main textbook.]
- Haykin, S. and Moher, M., Communication Systems, 5th edition, John Wiley, 2009.
- Carlson B. A. and Crilly P, Communication Systems, 5th edition, McGraw-Hill (Singapore), 2010.
- Proakis, J. G. and Salehi, M., Fundamentals of Communication Systems,

Prentice-Hall, 2004.
Learning outcomes: The students would be able to appreciate everyday communication technologies that we depend on, at a deeper level and can judiciously choose actions to further specialize in this dynamic and ever-changing technology.
Any other Remarks: There is a significant active learning in the course involving simulation exercises. **Course Objectives:** To understand the fundamental principles underlying the performance analysis, design, and optimization of analog and digital communication systems using analysis tools like Fourier transforms, energy/power spectral density for deterministic and random signals. Students will learn to relate the fundamental principles to prevalent communication systems of today like cell phones, Wifi, radio and TV broadcasting, satellites, and computer networks.